

Fluid Mechanics for Civil & Environmental Engineers

CIVL 2050 · Fall 2026 · 4 credits · Carnegie Building 113

A one-page visual semester map is the last page of this syllabus.

Course Information

Course	CIVL 2050 – Fluid Mechanics for Civil & Environmental Engineers
Sections	01, 02, 03
Lecture	M/R 10:00–11:50 AM; Carnegie Building 113
Labs	s01 Tuesday 10:00–11:50 AM; s02 Wednesday 10:00–11:50 AM; s03 Wednesday 2:00–3:50 PM
Website	https://lms.rpi.edu/
Prerequisites	ENGR 1100 (IEA) and PHYS 1100 (Physics I)
Final exam	TBA until Registrar final-exam assignment

Instructional Team and Office Hours

Role	Name / Email	Office	Office hours
Instructor	Dr. Daniel C. Lander landed2@rpi.edu	JEC 4030	Wed 2:00–4:30 PM JROWL 2C13
Teaching Assistant	Anu Aryal aryala@rpi.edu	JEC 5304	Tue 9:00 AM–12:00 PM
Teaching Assistant	Md Akib Arafin arafim@rpi.edu	JEC 5318	Mon 12:00–3:00 PM
Teaching Assistant	Jing Zhang zhangj47@rpi.edu	JEC 5308	Thu 12:00–3:00 PM

Course Text and Additional Resources

Text	<i>Fluid Mechanics for Engineers</i> , David A. Chin, Pearson, 2017/2022, 1st edition or any compatible edition, ISBN 9780137848317.
Additional resources	Course notes, lecture slides, laboratory handouts, report templates, homework statements, solution notes, and announcements will be posted through LMS and/or Gradescope. Some assignments may require tools such as Microsoft Excel, MATLAB, or water-distribution modeling software; submit only requested results, tables, figures, and written work unless otherwise instructed.

Course Purpose and Outcomes

CIVL 2050 is the introductory course in fluid and flow phenomena applicable to civil and environmental engineering problems. It is also the laboratory course in fluids and flow phenomena. The course emphasizes real-fluid behavior, engineering analysis, experimental measurement, and interpretation of observations in the context of theory.

Students who successfully complete this course shall be able to:

1. Analyze pressures exerted by fluids at rest and in uniform flow.
2. Apply conservation of mass, energy, and momentum to a control volume.
3. Analyze energy losses in pipe flows.
4. Use dimensionless numbers to evaluate pipe friction, pump performance, and model/prototype relationships.
5. Acquire skills in analyzing and solving engineering problems.
6. Make measurements of flow phenomena.
7. Apply hydrostatics, conservation of mass, energy, and momentum to control-volume problems.
8. Write succinct reports on experiments and discuss observations in the context of theory.

Course Assessment Measures

Assessment	Due date / timing	Learning outcome numbers
Exams	As specified below	1, 2, 3, 4, 7
Laboratories	As specified below	1, 2, 3, 4, 5, 6, 7, 8
Homework	As specified below	1, 2, 3, 4, 7

Course Organization and Assessment

The course is structured around lecture, laboratory experiments, homework, attendance, and examinations. Lecture meetings are two 110-minute meetings per week. Laboratory experiments are conducted in small teams during the assigned lab section.

Attendance Policy

Attendance is mandatory for this course. Because consistent presence is essential for learning and participation, attendance will count toward your final grade.

- Attendance will be recorded using the Blackboard Attendance tool.
- You may miss up to three classes without penalty during the semester. These absences are intended to cover minor illnesses, travel, or other personal circumstances. No documentation is required for the first three absences.
- For any additional absence, notify the instructor in advance when possible (e.g., for university travel or religious observances) or provide an explanation during office hours for consideration.
- Attendance is worth 5% of the total course grade.

Grading of Attendance:

- Exam 1 and Exam 2 are exempt from the attendance calculation. All other scheduled class meetings count unless the instructor exempts a meeting.
- In Blackboard, an “Excused” status applies to an individual student and does not count as an absence. An exempt meeting is excluded from the attendance calculation for the entire class.
- A “Late” status receives 50% attendance credit for that meeting.
- After the first three absences without penalty, each additional unexcused absence reduces the attendance grade proportionally across the remaining counted meetings.

Exams

Three closed-book exams are scheduled: Exam 1: Mon Sep 28, class time; Exam 2: Thu Nov 19, class time; Final Exam: TBA until Registrar final-exam assignment. Make up exams will be given at the discretion of the instructor.

Exam Crib Sheets. Crib sheet(s) are permitted in exams with the following specifications:

- Exam 1: one single-sided crib sheet. Exam 2: Exam 1 crib sheet plus a second single-sided crib sheet.
- Each is single sided, hand-written (no tablet) by the student, and includes name and RIN.
- Allowed: equations, brief notes, basic diagrams.
- Disallowed: full solutions to problems and extensive sections of course notes.

Disability Support Services Exams. Students with exam accommodations must follow the procedures below, consistent with RPI Testing Center policy:

- Students must provide the instructor with a current accommodation letter issued by RPI Access & Neurodiverse Student Success as soon as possible.
- Students are responsible for scheduling their own exams through the RPI Testing Center scheduling system. Time slots are limited and fill quickly, so students are encouraged to schedule as soon as possible based on the exam dates provided above/below. Book using the Testing Center request system.
- For this course, DSS exams must be taken on the same calendar day as the scheduled in-class exam, with the exam starting between 8:00 AM and 12:00 PM, unless explicitly approved in writing by the instructor.
- All exam requests require instructor approval through the Testing Center system. An exam appointment is not confirmed until it has been approved by both the instructor and the Testing Center.

Additional information regarding Testing Center policies and procedures is available at: <https://provost.rpi.edu/faculty-resources/testing-center>. Failure to submit appropriate documentation or to schedule an exam within the allowed time window may result in the exam being taken without accommodations.

Homework

Ten homework assignments will be assigned. Submit as a single PDF through Gradescope, reached from the course Blackboard site. Assignments are to be scanned using a dedicated scanning app; individual photos will not be accepted. Students are responsible for ensuring scan quality and legibility. The lowest homework grade will be dropped, excluding HW 10.

Homework is normally due Wednesday at 11:59 PM. The homework immediately before an exam is an exception: Friday night before a Monday exam, Monday night before a Thursday exam. Posted dates in the table below are approximate; the due date governs.

HW	Posted	Due (11:59 PM)	Anchor classes	Topics	Drop?
1	Thu Aug 27	Wed Sep 9	C01-C02	Ideal gas law, pressure, manometry	yes
2	Thu Sep 3	Wed Sep 16	C03-C04	Forces on plane and curved surfaces, buoyancy	yes
3	Mon Sep 14	Wed Sep 23	C05-C06	Flow kinematics, conservation of mass	yes
4	Thu Sep 17	Fri Sep 25	C07-C08	Conservation of energy, Bernoulli equation	yes
5	Mon Oct 5	Wed Oct 14	C11-C12	Flow measurement; conservation of momentum	yes
6	Thu Oct 15	Wed Oct 21	C13-C14	Dimensional analysis and similitude	yes
7	Mon Oct 19	Wed Oct 28	C15-C16	Laminar and turbulent flow in pipes	yes
8	Mon Oct 26	Wed Nov 11	C17-C19	Steady-flow energy equation; pipe systems	yes
9	Thu Nov 5	Mon Nov 16	C20-C22	Pump performance; boundary layers and drag	yes
10	Mon Nov 30	Wed Dec 9	C25-C26	Turbines and hydropower	no

Homework Presentation

Layout specifications:

1. Include a clear header on the first page with: name, homework number, date, and page numbers. Number pages in the format A/B (e.g., 2/6).
2. Use ruled engineering paper (physical or digital). Start each problem on a new page.
3. Show work in a logical order so another engineer can follow it. Use this structure: given data / diagram; required results; separator line; calculations / diagram; final answer (boxed).
4. Show steps with no big jumps. For repetitive calculations, one fully worked sample plus a table is acceptable.
5. Box final answers and include correct units on intermediate results and final results.

Laboratory Experiments

Students will complete three laboratory experiments designed to develop proficiency with the experimental method: observation, modeling, prediction, and evaluation of fluid behavior. A theoretical model is used to predict experimental outcomes and is compared with measured results. A formal report for each experiment is required. **All three lab reports must be submitted to pass the course.**

Lab execution is scheduled by section; report due dates and pre-lab submission dates are listed separately because the writing and analysis burden is the main scheduling constraint.

Experiment	Sec.	Date	Hour 1 (A/B)	Hour 2 (C/D)	Pre-lab	Report
Lab 1 Flow meters	01	Tue Oct 6	10:00–11:00 AM	11:00–11:50 AM	Mon Oct 5	Sun Oct 18
	02	Wed Oct 7	10:00–11:00 AM	11:00–11:50 AM	Tue Oct 6	Sun Oct 18
	03	Wed Oct 7	2:00–3:00 PM	3:00–3:50 PM	Tue Oct 6	Sun Oct 18
Lab 2 Impact of a jet	01	Tue Oct 20	10:00–11:00 AM	11:00–11:50 AM	Mon Oct 19	Sun Nov 1
	02	Wed Oct 21	10:00–11:00 AM	11:00–11:50 AM	Tue Oct 20	Sun Nov 1
	03	Wed Oct 21	2:00–3:00 PM	3:00–3:50 PM	Tue Oct 20	Sun Nov 1
Lab 3 Energy loss in pipes + Osborne Reynolds	01	Tue Nov 10	10:00–11:00 AM	11:00–11:50 AM	Mon Nov 9	Sun Dec 6
	02	Wed Nov 11	10:00–11:00 AM	11:00–11:50 AM	Tue Nov 10	Sun Dec 6
	03	Wed Nov 11	2:00–3:00 PM	3:00–3:50 PM	Tue Nov 10	Sun Dec 6

Each section meets only on the three dates shown. Pre-lab and report submissions are due at 11:59 PM.

Grading Criteria and Submission Policies

Component	Option 1 (with final)	Option 2 (no final)
Homework ¹	5%	5%
Attendance ²	5%	5%
Exam 1 ³	20%	35%
Exam 2 ³	20%	35%
Labs (3) ⁴	20%	20%
Final exam	30%	0%
Total	100%	100%

¹ lowest homework grade dropped; HW 10 is not drop-eligible. ² three unexcused absences are free; see Attendance Policy. ³ RCA bonus possible; credit calculated on the Option 1 distribution. ⁴ all three lab reports must be submitted to pass the course.

Regrade Requests. All grading re-evaluations (excluding exams) will be handled through Gradescope's regrade request system. Students must submit any regrade requests within one week of the grade being posted. After this one-week window, no regrade requests will be considered, regardless of the circumstances. It is the student's responsibility to review feedback promptly and submit requests within the stated timeframe. Exam re-evaluations must be made during office hours within two weeks of the return date.

Late Submissions. Late assignments will be handled as follows:

Time (t) after deadline	Deduction
$t \leq 1$ hour	10%
$1 < t \leq 48$ hours	20%
$48 \text{ hours} < t \leq 7$ days	50%

Letter Grades and Rounding

Typical letter grades for the cumulative score at the end of the course are shown below. These are minimum expectations and, if overall class performance exceeds the instructor's standards, the criteria may be relaxed.

Letter	Range	Letter	Range	Letter	Range
A	93–100	B-	80–82.99	D+	67–69.99
A-	90–92.99	C+	77–79.99	D	63–66.99
B+	87–89.99	C	73–76.99	D-	60–62.99
B	83–86.99	C-	70–72.99	F	0–59.99

Final Grade Rounding. Final course grades will be calculated according to the grading scale listed above. Grades will be rounded up to the nearest tenth of a percent (0.1%). For example, a grade of 92.91% will be recorded as 93.0% (A). No additional considerations (e.g., participation, effort, or “close calls”) will be applied beyond this rounding policy.

Opportunities for Extra Credit. After the two exams, students may submit a Root Cause Analysis (RCA) to analyze their exam performance. The RCA may redeem 5 points, weighted according to Option 1 grade distributions, on the original exam score. One point (1%) of bonus credit is also available if final course evaluations are completed to 90% or above for the whole class (all sections).

Laptops and Mobile Phones. To promote attention and reduce distractions, the use of laptops and mobile phones is not permitted during class, unless specifically authorized by the instructor for a class activity. Tablets are permitted for note-taking and course-related use. All other devices must be silenced and put away after attendance is taken. Students with documented accommodations may use devices as required. In case of emergency, notify the instructor and step outside if a phone is needed. Unauthorized use of laptops or phones may result in being marked absent.

Course Schedule

Wk	Date	Class	Topic / activity	Reading	Notes
1	Thu Aug 27	C01	Introduction and overview; units; fluid properties	1.1-1.10	First class of the term
2	Mon Aug 31 Thu Sep 3	C02 C03	Pressure; manometry Forces on plane surfaces	2.2-2.3 2.4	
3	Mon Sep 7 Thu Sep 10	– C04	No class — Labor Day Forces on curved surfaces; buoyancy	– 2.5-2.6	Follows Monday schedule
4	Mon Sep 14 Thu Sep 17	C05 C06	Classifications of flows; systems and control volume; fluid kinematics Reynolds transport theorem; conservation of mass	3.1, 3.2, 4.1 4.1-4.3	
5	Mon Sep 21 Thu Sep 24	C07 C08	The Bernoulli equation (conservation of energy) Exam 1 review	3.3 course notes	No new exam content
6	Mon Sep 28 Thu Oct 1	C09 C10	Exam 1 Estimation of uncertainty and lab reports	– course notes	covers C01-C08 Placed immediately after Exam 1
7	Mon Oct 5 Thu Oct 8	C11 C12	Exam 1 solutions; flow measurement Conservation of linear momentum	3.4 4.4	
8	Mon Oct 12 Thu Oct 15 Fri Oct 16	– C13 C14	No class — Columbus Day / Fall Break Dimensional analysis Similitude	– 6.1-6.5 6.6	Follows Monday schedule
9	Mon Oct 19 Thu Oct 22	C15 C16	Viscosity; laminar flow in pipes Turbulent flow in pipes	1.6, 7.1-7.3 7.4	
10	Mon Oct 26 Thu Oct 29	C17 C18	Steady-flow energy equation and minor losses Single-pipe flow problems	course notes, 7.5.6 7.5.1-7.5.4	
11	Mon Nov 2 Thu Nov 5	C19 C20	Multiple-pipe problems Pump performance	course notes 8.3.1-8.3.3, 8.3.5-8.3.6	
12	Mon Nov 9 Thu Nov 12	C21 C22	Boundary layers and drag Flow over circular cylinders and spheres	10.1, 10.2, 10.3.1, 10.3.5-10.3.7 10.3.2	Movable after Thanksgiving if needed
13	Mon Nov 16 Thu Nov 19	C23 C24	Exam 2 review Exam 2	course notes –	No new exam content covers C10-C23
14	Mon Nov 23 – Fri Nov 27	–	No class — Thanksgiving Break	–	
15	Mon Nov 30 Thu Dec 3	C25 C26	Exam 2 solutions; hydraulic turbines and hydropower Hydraulic turbines and hydropower	8.5 8.5	
16	Mon Dec 7 Thu Dec 10	C27 C28	Course synthesis and problem-solving buffer Final exam review	course notes course notes	Last CIVL 2050 class meeting

Final exam date, time, and room are set by the Registrar and released mid-semester; the final-exam period is Dec 16–22. Thu Dec 10 is Class 28 and the final scheduled CIVL 2050 meeting.

Assessment Calendar

Every submission and action date in one place.

Date / time	Item	Type	Notes
Wed Sep 9, 11:59 PM	HW 1 due	Homework	
Wed Sep 16, 11:59 PM	HW 2 due	Homework	
Wed Sep 23, 11:59 PM	HW 3 due	Homework	Due before review/exam week pressure peaks
Fri Sep 25, 11:59 PM	HW 4 due	Homework	Exam-squeeze exception before Monday Exam 1
Mon Sep 28, class time	Exam 1	Exam	covers C01-C08
Mon Oct 5, 11:59 PM	Lab 1 pre-lab due (01)	Pre-lab	night before the Tue lab
Tue Oct 6, 11:59 PM	Lab 1 pre-lab due (02/03)	Pre-lab	night before the Wed lab
Wed Oct 14, 11:59 PM	HW 5 due	Homework	
Sun Oct 18, 11:59 PM	Lab 1 report due	Lab report	Flow meters
Mon Oct 19, 11:59 PM	Lab 2 pre-lab due (01)	Pre-lab	night before the Tue lab
Tue Oct 20, 11:59 PM	Lab 2 pre-lab due (02/03)	Pre-lab	night before the Wed lab
Wed Oct 21, 11:59 PM	HW 6 due	Homework	
Wed Oct 28, 11:59 PM	HW 7 due	Homework	
Sun Nov 1, 11:59 PM	Lab 2 report due	Lab report	Impact of a jet
Mon Nov 9, 11:59 PM	Lab 3 pre-lab due (01)	Pre-lab	night before the Tue lab
Tue Nov 10, 11:59 PM	Lab 3 pre-lab due (02/03)	Pre-lab	night before the Wed lab
Wed Nov 11, 11:59 PM	HW 8 due	Homework	before Exam 2
Mon Nov 16, 11:59 PM	HW 9 due	Homework	Exam-squeeze exception before Thursday Exam 2
Thu Nov 19, class time	Exam 2	Exam	covers C10-C23
Mon Nov 23	Thanksgiving Break	No class	Nov 23–27; protected grading / catch-up buffer
Sun Dec 6, 11:59 PM	Lab 3 report due	Lab report	Energy loss in pipes + Osborne Reynolds
Wed Dec 9, 11:59 PM	HW 10 due	Homework	not drop-eligible; not drop-eligible
TBA (Dec 16–22)	Final exam	Exam	cumulative; weighted toward post-Exam-2 content (C25-C26)

Inclusivity, Accessibility, Title IX, and Wellbeing

Inclusivity and Accessibility. The instructional team strives to create an equitable and inclusive learning environment that ensures the full participation and success of all students in the course. We expect all class members to be respectful of diverse perspectives and individuals, and to use inclusive language in both written and oral work. We recognize that students learn in different ways, and we encourage students to communicate with the instructional team about individual learning needs and how this course can best support them.

Students with a documented disability, or who believe they may require accommodations, should contact Access and Neurodiverse Student Success (ANSS) as early as possible. Academic accommodations require official registration with ANSS. Once registered, students should promptly notify the instructor so that accommodations can be implemented in a timely manner. ANSS contact information: dss@rpi.edu; 518-276-8197.

Title IX. Rensselaer Polytechnic Institute recognizes that to maintain a healthy, safe, and vibrant living and learning community, it must continue to foster an environment free from gender inequality and sexual violence. RPI's Title IX office administers Title IX policies and thoroughly investigates all Title IX complaints. [Title IX resources](#) are available for all members of the campus community.

Student Wellbeing & RPI Campus Resources. All students' holistic wellbeing, including physical, mental, and emotional health, is of importance throughout the semester. Good health and wellbeing practices contribute to overall academic success. Feel free to communicate with your instructor throughout the semester about your wellbeing, especially if you are facing wellbeing or health challenges that may affect your attendance and performance in the course and/or need assistance getting in touch with campus academic support and wellbeing resources.

Resource	Location / link	Contact
Student Success Center	First floor of Folsom Library	contact info to come
Engineering Academic HUB & Faculty Advisors	First-Year Advising Hub	–
Dean of Students Office	doso.rpi.edu	(518) 276-6266
Counseling Center	Counseling Center	(518) 276-6479
Mantra Health	Mantra Health Info / Mantra Hub	ConnectNow support: (518) 276-8888
Student Health Center	Student Health Center	(518) 276-6287
Rec Well at the Mueller Center	Rec Well	–

Academic Integrity

This statement is intended to cover most, but not all, forms of academic dishonesty. Please refer to The Rensselaer Handbook of Student Rights and Responsibilities for detailed information on Institute policy in this regard.

Generative AI and AI-Assisted Technologies. Regarding the use of Generative AI and AI-assisted technologies by students, the course policy is as follows:

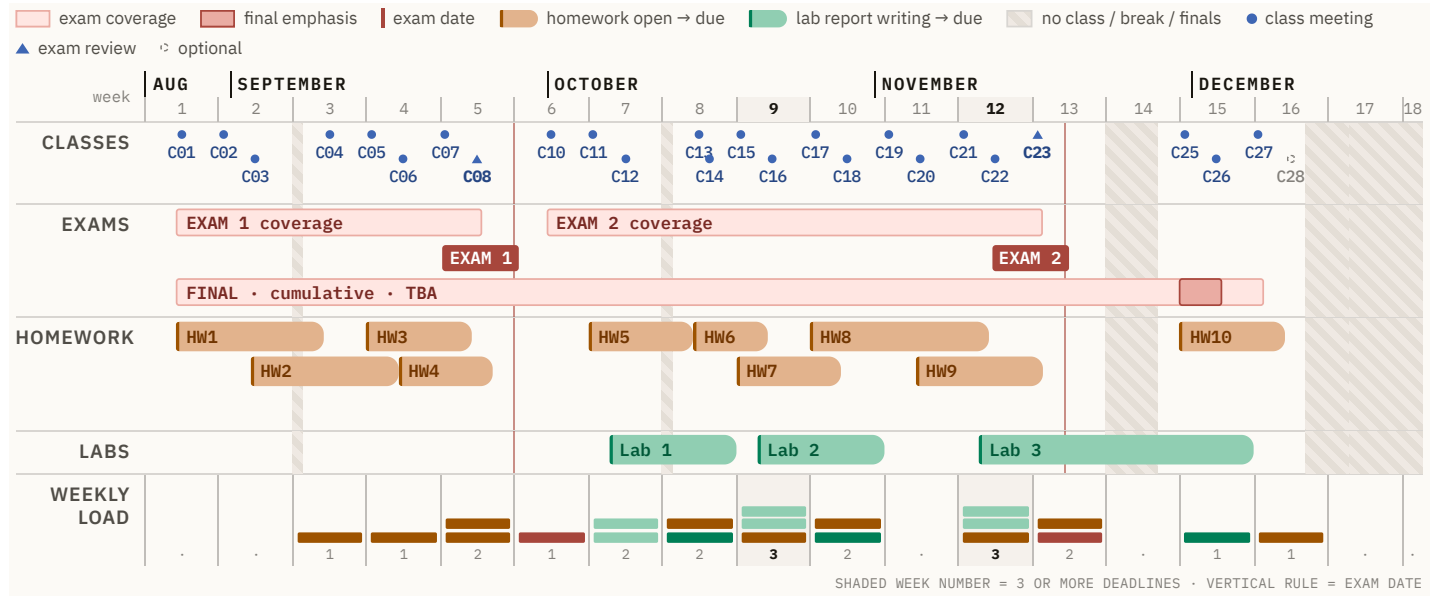
- Only use these technologies to improve readability and language, not to replace key student learning tasks such as interpreting data or drawing analysis conclusions. Apply the technology with human oversight and control, and carefully review and edit the result, as AI can generate authoritative-sounding output that can be incorrect, incomplete, or biased.
- Disclose in submissions the use of AI and AI-assisted technologies in the working process. Students are ultimately responsible and accountable for the contents of their submissions and must therefore be able to demonstrate that material generated has been cross-checked and verified from a different source/reference.

Homework and Labs. Collaboration among students in the class is permitted and encouraged. Seeking help from other sources

is also encouraged. However, the work you submit should be your own. Students found to have copied any part of their homework from a solution manual or from an online database will be considered in breach of academic honesty. Appropriate recourse will sort based on the circumstances.

Exams. Collaboration during an exam (exchanging verbal or written information or copying) is prohibited. In addition, use of email or the internet during exams for any purpose is prohibited. All mobile devices (smart phones, computers, etc.) must be stored securely away during exams and are not to be used unless specifically directed otherwise by the instructor. Use of (or ANY interaction with) a mobile device during an exam without explicit permission of the instructor will be interpreted as the illicit transfer of exam data, will be considered an act of cheating, and will be treated as such. All parties involved will receive an F in the course.

Semester Timeline



CIVL 2050 • Fluid Mechanics

Semester map — lectures, homework, labs, and exams, week by week.

FALL 2026
AUG 27 – DEC 22

	MON	TUE	WED	THU	FRI	S	S
W1 8/24–8/30	24	25	26	27 • C01	28	29	30
				HW1			
W2 8/31–9/6	31 • C02	1	2	3 • C03	4	5	6
				HW2			
W3 9/7–9/13	7 No class	8	9 HW1	10 • C04	11 Mon sched	12	13
				HW3			
W4 9/14–9/20	14 • C05	15	16 HW2	17 • C06	18	19	20
		HW4					
W5 9/21–9/27	21 • C07	22	23 HW3	24 • C08	25 HW4	26	27
W6 9/28–10/4	28 EXAM 1	29	30	1 • C10	2	3	4
				HW5			
W7 10/5–10/11	5 • C11 PL1 01	6 • Lab1 01 PL1 02-03	7 • Lab1 02-03	8 • C12	9	10	11
			R1		HW6		
W8 10/12–10/18	12 No class	13	14 HW5	15 • C13	16 • C14	17 Mon sched	18 R1
W9 10/19–10/25	19 • C15 PL2 01	20 • Lab2 01 PL2 02-03	21 • Lab2 02-03 HW6	22 • C16	23	24	25
	HW7		R2				
W10 10/26–11/1	26 • C17	27	28 HW7	29 • C18	30	31	1 R2
W11 11/2–11/8	2 • C19	3	4	5 • C20	6	7	8
				HW9			
W12 11/9–11/15	9 • C21 PL3 01	10 • Lab3 01 PL3 02-03	11 • Lab3 02-03 HW8	12 • C22	13	14	15
			R3				
W13 11/16–11/22	16 • C23 HW9	17	18	19 EXAM 2	20	21	22
W14 11/23–11/29	23 TG	24 TG	25 TG	26 TG	27 TG	28	29
W15 11/30–12/6	30 • C25	1	2	3 • C26	4	5	6 R3
		HW10					
W16 12/7–12/13	7 • C27	8	9 HW10	10 • C28	11	12	13
W17–18 12/14–12/22	STUDY PERIOD DEC 12–15 FINAL EXAM PERIOD DEC 16–22 – DATE TBA						